

L01 Exploring Real-World Applications of Computer Vision

1. Description of the Application

With experience in physical security, I have seen how conventional surveillance depends largely on stationary CCTV cameras that inevitably possess blind spots, as well as foot patrol staffs that are susceptible to fatigue, bad weather conditions, and physical danger. Mobile security patrol robots, including the outdoor Knightscope K5 robot and the indoor Cobalt Robotics robot, solve the problem of conventional surveillance inefficiencies by offering autonomous mobile surveillance around 360 degrees. These mobile robots operate within commercial premises, corporate complexes, parking garages, and transit stations. The main function of mobile robots in surveillance is being mobile guards that move along specified paths and alert about the presence of safety threats like fires, blocked emergency exits, or water leaks, about any people where they should not be, and alerting on-duty guards of the incidents via real time communication.

The implementation of security robotics is being scaled up in the surrounding areas of Austin where I just moved from. Northeast Austin communities have deployed community patrol robots, for example, Palmer, to help deter thefts and notify the community of any anomalies at the periphery of their surroundings. The Austin City Council has authorized the deployment of quadruped robotic sensing units, like the Boston Dynamics' Spot robot, in the Austin Police Department for doing remote surveillance and analysis of dangerous and barricaded situations without putting the lives of human officers at risk. Besides, commercial facilities and transit perimeters in Central Texas deploy autonomous mobile units for securing parking facilities.

2. Technology Behind It

The basic features of the autonomous security patrol robot depend upon the sensory pipeline that uses cameras, depth sensors, and deep learning techniques to analyze the environment around the robot. 4K optical cameras take videos continuously from all sides of the robot, while the thermal and long-wave infrared (LWIR) cameras detect any heat variance so that the presence of intruders in zero light can be detected or heating of electrical equipment can be detected. To add 3D information to the 2D imaging, LiDAR and ultrasonic sonar sensors are used.

From an algorithmic perspective, the robot makes its way through unstructured spaces employing Visual Simultaneous Localization and Mapping (Visual SLAM). According to Durrant-Whyte & Bailey (2006), the basic concept of this technique can be described as "the problem of building a map of an unknown environment by a mobile robot while at the same time navigating the environment using this map" (p. 99). For the purpose of visual data processing in real-time, the robot utilizes single-stage convolutional neural networks, namely YOLO architecture. According to Redmon et al. (2016), the main characteristic of the YOLO framework consists in its approach to object detection, which can be described as "framing the object detection problem as a single regression problem, straight from image pixels to bounding box

coordinates and class probabilities" (p. 779), thus allowing for processing "streaming video in real-time with less than 25 milliseconds of latency" (p. 780). This allows the robot to distinguish people, vehicles, bicycles, and loose objects instantly. Moreover, the robot uses optical character recognition (OCR) along with deep learning methods for Automatic License Plate Recognition (ALPR). He & Zhang (2023) state that combining automated vision modules into mobile devices allows the robot to "realize real-time image acquisition, target detection, and license plate recognition" (p. 95) in order to identify the authorized access. In his turn, Szeliski (2022) stresses that the core of robust robotic vision systems lies in background subtraction and motion estimation because "detecting moving objects in video streams is a fundamental low-level task in computer vision" (p. 615).

3. Benefits and Challenges

Based on my observations during security shifts, the main advantages of autonomous patrol robots are numerous. First of all, an undeniable benefit of the autonomous patrol robots lies in their capacity to remain active throughout 24 hours in a row. Second, the most important feature of the robot is its capability to act as a safety buffer; for instance, in case of suspicion of burglary or chemical spills, a robot can be sent out to assess the situation and not put the guards at risk physically.

Yet, there are some particular practical, technical, and ethical limitations with regards to its use. First of all, extreme weather conditions, such as rainstorms, fog, lens splashes with mud, or even direct sunlight may affect optical quality and result in failures or false alerts. In addition, ordinary wheeled systems are restricted by terrain and cannot go up and down stairs or jump over high curbs. From the ethical point of view, constant filming of high-quality videos in semi-public areas poses serious problems of surveillance, biometrics data storage, and algorithms' biases in detecting anomalies.

4. Reflection

In terms of the future, the autonomous security robots will be getting more intelligent by way of the use of multimodal foundation models in edge-computing. Future robots will no longer depend exclusively on bounding boxes as the sole input; they will be able to comprehend complicated scene context and understand natural languages for communicating directly with distraught people. As far as I'm concerned, being a security guard, future robots will not take the place of human officers. Instead, they will be invaluable force multipliers, performing automated rounds while humans concentrate on de-escalation, coordination, and assisting tenants.

Society stands to gain immensely from the use of this technology in terms of securing common areas, transportation centers, and business campuses, with minimal risk of humans being affected by acts of violence. However, there are negative implications that arise from the use of this technology in society in the absence of any governing framework.

References

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